Full Stack Assignment – 17

Assignment ID: FSD-17

Title: Approval Workflow Management System

Duration: 3 Hours

1️⃣ Objective

Design and develop an **Approval Workflow Management System** that allows users to create requests and approvers to review them.
The system must enforce approval ownership rules and ensure **data persistence using a database**.

2️⃣ User Roles

CREATOR

- Register and log in to the system
- Create approval requests
- View own request status

APPROVER

- Log in to the system
- View all pending approval requests
- Approve or reject requests

3️⃣ Functional Requirements

Authentication & Authorization

- Implement user registration and login
- Use **JWT-based authentication**
- Protect all backend APIs using JWT
- Enforce **role-based access control** on APIs and frontend views

Request Management

- CREATOR can submit requests with required details
- Requests must be stored in the database
- APPROVER can approve or reject requests
- Request status must be retrieved dynamically

4️⃣ Business Rule (Critical)

⚠️ A creator must not be allowed to approve their own request.

Approval attempts by the same user who created the request must be rejected.

5️⃣ Technical Constraints (Mandatory)

- All request and approval data **must persist in a database**
- Hardcoded or in-memory storage is **not allowed**
- Backend APIs must reflect real database state
- JWT authentication and role validation are mandatory
- Use environment variables for sensitive configuration

6️⃣ Frontend Requirements

- Login and registration pages
- Role-based dashboards for CREATOR and APPROVER
- Request creation form
- Request listing with approval actions

UI design is not evaluated; rule enforcement and flow matter.

7️⃣ Submission Requirements (Strict)

GitHub Repository

- Backend source code
- Frontend source code
- Proper folder structure
- Clear and meaningful commit history

Live Deployment

- Live frontend deployment link
- Live backend deployment link
- Backend must be connected to the database

README.md (Mandatory)

The README file must include:

- Project overview
- Tech stack used
- User roles and permissions
- API endpoints
- Database schema or collections

- Steps to run the project locally
- Live frontend and backend links

8️⃣ Evaluation Note

Evaluation focuses on **authorization logic, rule enforcement, and end-to-end integration**.